

Chapter 2

Frequency, Probability, and Distributions

This chapter introduces several of the basic tools used to organize and interpret data. We begin with the distinction between a population and a sample, then move to frequency tables, relative frequencies, bar charts, histograms, and probability distributions.

These ideas are closely connected. Frequency tells us how often an outcome appears in a data set. Relative frequency converts those counts into proportions. Probability formalizes the idea of likelihood, and a probability distribution records the probabilities assigned to all possible outcomes. The chapter therefore moves from observed data to the beginnings of statistical description.

2.1 Populations and Samples

Statistical work usually begins with a practical limitation: we want information about a large group, but we can only observe part of it directly. This leads to the distinction between a population and a sample. The population is the full group of interest, while the sample is the part that is actually observed.

Definition. *Population.* A **population** is the complete collection of individuals or items under study.

Definition. *Sample.* A **sample** is a subset of the population that is actually observed.

In many situations it is not realistic to observe every member of the population. A carefully chosen sample makes the problem manageable while still allowing useful conclusions about the larger group.

Definition. *Simple random sample.* A **simple random sample** is one in which every member of the population has an equal probability of being selected.

If a sample is selected in a way that favors some members of the population over others, then the sample is **biased**. In that case, the data may give a misleading picture of the whole population.

Example 2.1.1. A school administrator wants to know whether students prefer longer lunch periods. She surveys every student in one advisory class.

The population is all students in the school. The sample is the students in that one advisory class.

This sample may be biased if the advisory class is not representative of the whole school. For example, if the class consists entirely of seniors, the preferences of younger students are not captured.

Example 2.1.2. A biologist wants to estimate the average weight of fish in a lake. She catches 50 fish at random locations across the lake and weighs each one.

The population is all fish in the lake. The sample is the 50 fish she caught.

If she only fished near one bank, the sample would be biased toward fish that prefer that habitat.

Exercises 2.1

2.1.1. For each situation, identify the population and the sample. State whether the sampling method is likely to produce a biased result, and explain why.

- (a) A hospital surveys patients who visited the emergency room last Tuesday to estimate the average wait time across the whole month.
- (b) A polling organization calls 1,000 randomly selected registered voters to estimate support for a ballot measure.
- (c) A coffee shop owner asks customers who come in on Monday morning whether they would like the shop to open earlier.

2.1.2. A researcher wants to estimate the average screen time per day for teenagers in a city. Describe a sampling method that would produce a simple random sample. Describe a method that would produce a biased sample.

2.1.3. Explain in two complete sentences why a larger sample does not automatically correct for bias.

2.2 Frequency Tables

Once data has been collected, it must be organized before it can be interpreted. One of the simplest ways to do this is to count how often each outcome occurs. Frequency tables provide a clear summary of those counts.

Definition. *Frequency.* The **frequency** of an outcome is the number of times that outcome appears in the data set.

Definition. *Frequency table.* A **frequency table** lists each distinct outcome together with its frequency.

Because every observation contributes to exactly one count, the sum of the frequencies must equal the total number of observations in the data set.

Example 2.2.1. Twenty-five students were asked to name their preferred study location. The responses were:

library, home, library, cafe, home, library, home, library, cafe, library,
home, library, home, cafe, library, home, library, cafe, home, library,
library, home, cafe, library, home

Counting each response:

Location	Frequency
Library	11
Home	9
Cafe	5
Total	25

The most common response is Library, with 11 out of 25 students.

Example 2.2.2. (*Numerical data*) A quality inspector records the number of defects found in each of 20 products examined. The results are: 0, 1, 0, 2, 1, 0, 1, 3, 0, 1, 2, 0, 1, 0, 1, 0, 2, 1, 0, 1.

Defects	Frequency
0	8
1	8
2	3
3	1
Total	20

Exercises 2.2

2.2.1. The following data records the number of siblings reported by each of 18 students:

2, 0, 1, 3, 1, 2, 0, 1, 1, 2, 3, 0, 1, 2, 1, 0, 2, 1.

- (a) Construct a frequency table.
- (b) What is the most common number of siblings?
- (c) Verify that the frequencies sum to 18.

2.2.2. The following data records the transportation method used by 20 employees to commute to work: car, bus, car, bike, car, walk, car, bus, bike, car, car, walk, bus, car, bike, car, bus, car, walk, car.

- (a) Construct a frequency table.
- (b) Which method is most common?
- (c) How many employees use something other than a car?

2.2.3. A frequency table for a data set with four categories shows frequencies of 8, 12, and 5 for the first three categories. If the total number of observations is 30, what is the frequency of the fourth category?

2.3 Relative Frequency and Probability

Raw counts are useful, but counts alone do not always make different data sets easy to compare. Relative frequency solves this problem by converting counts into proportions. Once we do that, the connection to probability becomes much clearer.

Definition. *Relative frequency.* The **relative frequency** of an outcome is its frequency divided by the total number of observations.

$$\text{Relative frequency} = \frac{\text{frequency}}{\text{total}}$$

Relative frequencies are numbers between 0 and 1, and they describe proportions rather than raw totals. Because they account for the entire data set, the relative frequencies in a table always add to 1.

Relative frequency also serves as a bridge to probability. When an experiment or process is repeated many times under similar conditions, the relative frequency of an outcome tends to settle near its probability.

Definition. Probability. The **probability** of an outcome is a number P between 0 and 1 measuring how likely that outcome is. $P = 0$ means the outcome is impossible. $P = 1$ means the outcome is certain.

Example 2.3.1. Adding a relative frequency column to the study location data from Example 2.2.1:

Location	Frequency	Fraction	Decimal
Library	11	$\frac{11}{25}$	0.44
Home	9	$\frac{9}{25}$	0.36
Cafe	5	$\frac{5}{25}$	0.20
Total	25	$\frac{25}{25}$	1.00

Based on this sample, the estimated probability that a randomly selected student prefers studying in a library is 0.44.

Example 2.3.2. A bag contains 4 red marbles, 3 blue marbles, and 5 green marbles. One marble is drawn at random. The probability of each outcome is:

$$P(\text{red}) = \frac{4}{12} = \frac{1}{3} \approx 0.333$$

$$P(\text{blue}) = \frac{3}{12} = \frac{1}{4} = 0.250$$

$$P(\text{green}) = \frac{5}{12} \approx 0.417$$

Check: $\frac{4}{12} + \frac{3}{12} + \frac{5}{12} = \frac{12}{12} = 1$. ✓

Example 2.3.3. (Complement) The **complement** of an event A is the event that A does not occur. Its probability is $P(\text{not } A) = 1 - P(A)$.

In the marble example:

$$P(\text{not red}) = 1 - P(\text{red}) = 1 - \frac{1}{3} = \frac{2}{3}.$$

This can be verified directly: $P(\text{blue}) + P(\text{green}) = \frac{3}{12} + \frac{5}{12} = \frac{8}{12} = \frac{2}{3}$.

Exercises 2.3

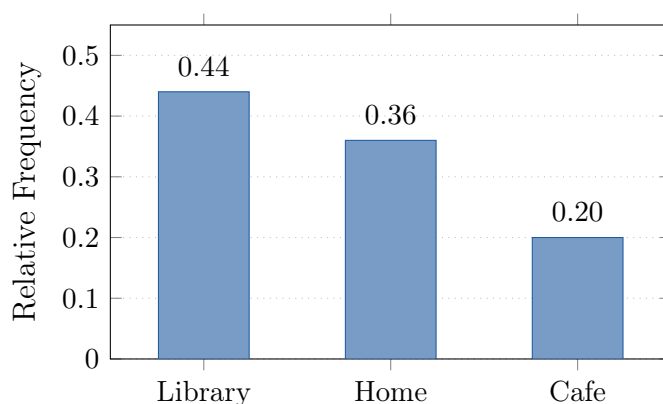
- 2.3.1. Using the sibling data from Exercises 2.2, problem 1:
- (a) Add a relative frequency column to your table. Express each as a fraction and as a decimal.
 - (b) Verify that the relative frequencies sum to 1.
 - (c) What is the estimated probability that a randomly selected student from this group has exactly one sibling?
 - (d) What is the probability of having more than one sibling?
- 2.3.2. A spinner is divided into five equal sections colored red, orange, yellow, green, and blue.
- (a) What is the probability of landing on any one color?
 - (b) What is the probability of not landing on red?
 - (c) What is the probability of landing on red or orange?
- 2.3.3. A student rolls a standard six-sided die.
- (a) Construct a probability distribution for all possible outcomes.
 - (b) What is $P(\text{rolling a 4})$?
 - (c) What is $P(\text{rolling an even number})$?
 - (d) What is $P(\text{rolling a number greater than 4})$?
- 2.3.4. A probability distribution has four outcomes with probabilities 0.15, 0.30, 0.25, and p . Find p and explain how you know.

2.4 Bar Charts

Tables organize information numerically, but graphs often make patterns easier to see. A **bar chart** displays a frequency or relative frequency table visually by assigning one bar to each category and using the height of the bar to represent the count or proportion.

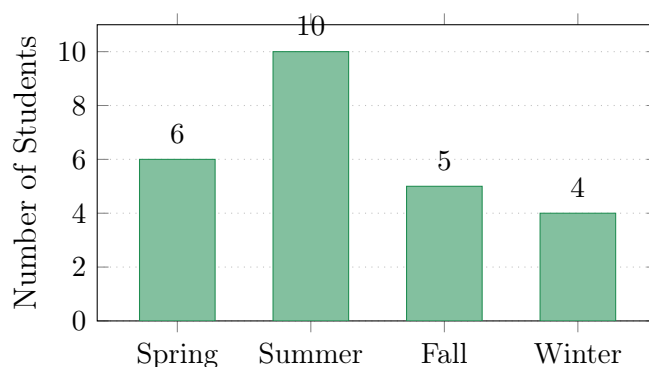
Bar charts are used for **categorical** data, where the outcomes are labels rather than measurements on a number line. The bars are separated from one another because the categories are distinct.

Example 2.4.1. Bar chart of the study location data, using relative frequencies.



The tallest bar identifies the most common outcome. Library has the highest relative frequency at 0.44.

Example 2.4.2. (*Reading a bar chart*) The bar chart below shows the distribution of favorite seasons for a group of students.



From this chart: 25 students were surveyed in total. Summer is the most popular season with 10 students. The probability that a randomly selected student from this group prefers Fall is $\frac{5}{25} = 0.20$.

Exercises 2.4

2.4.1. Using the transportation data from Exercises 2.2, problem 2:

- Compute the relative frequency for each transportation method.
- Draw a bar chart using relative frequencies on the vertical axis. Label each bar.
- Based on your chart, which two methods combined account for more than half of the commuters?

2.4.2. The bar chart below shows the number of books read last month by members

of a book club.



- How many members are in the book club?
- What is the relative frequency of members who read exactly 1 book?
- What is the probability that a randomly selected member read more than 2 books?

2.4.3. A survey of 40 office workers records their preferred lunch location: cafeteria (18), restaurant (10), desk (8), outside (4). Draw a bar chart using relative frequencies and identify the most and least common preferences.

2.5 Histograms

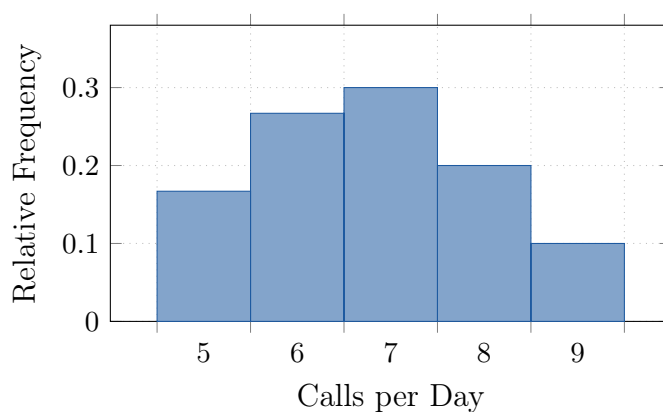
When the data is numerical rather than categorical, a histogram is usually more appropriate than a bar chart. A **histogram** shows how the data is distributed across values or intervals of values on a number line.

This makes histograms especially useful for seeing shape. They can show where data is concentrated, whether it is spread out or clustered, and whether the distribution appears roughly symmetric or uneven. Because the horizontal axis represents numerical values in order, the bars in a histogram touch.

Example 2.5.1. The number of daily customer service calls received by a small business over 30 days:

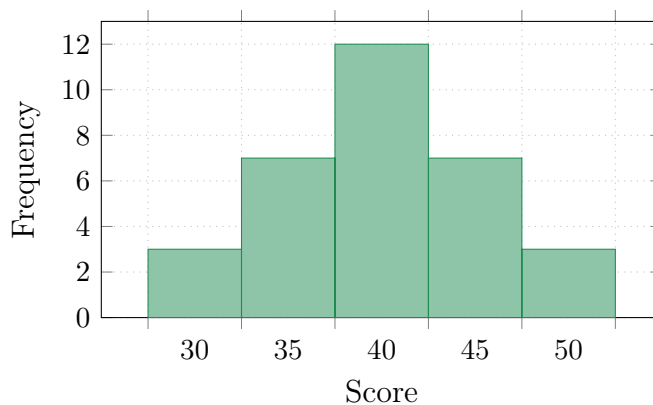
5, 8, 6, 7, 9, 5, 6, 8, 7, 6, 7, 8, 5, 9, 7, 6, 8, 7, 6, 5, 8, 7, 6, 9, 7, 8, 6, 7, 5, 8.

Calls	Frequency	Relative Frequency
5	5	$5/30 \approx 0.167$
6	8	$8/30 \approx 0.267$
7	9	$9/30 = 0.300$
8	6	$6/30 = 0.200$
9	3	$3/30 = 0.100$
Total	30	1.000



The distribution peaks at 7 calls per day. The most likely outcome for any given day is 7 calls.

Example 2.5.2. (*Grouped data*) When data spans a wide range, values are grouped into **intervals** called bins. The histogram below shows the distribution of scores on a 50-question quiz for a class of 32 students.



The distribution is symmetric: the bars rise to a peak at the 40–45 interval and fall symmetrically on both sides. This mound shape is encountered again in Chapter 4.

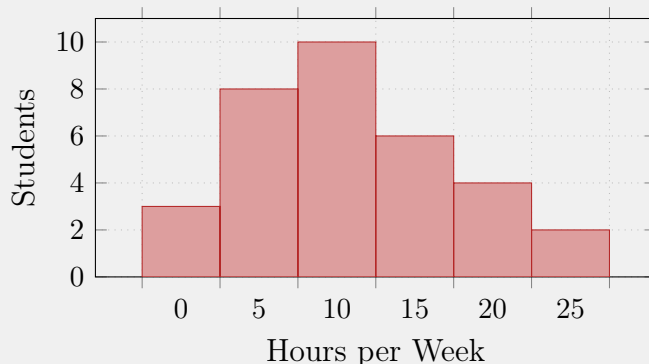
Exercises 2.5

2.5.1. The following data records the ages of 24 volunteers at a community event:

22, 25, 31, 28, 35, 22, 29, 40, 33, 27, 25, 38, 30, 26, 42, 23, 31, 29, 35, 24, 27, 33, 28, 36.

- Group the data into the bins $[20, 25)$, $[25, 30)$, $[30, 35)$, $[35, 40)$, $[40, 45)$. Construct a frequency table.
- Draw a histogram using relative frequencies.
- In which age range do the most volunteers fall?

2.5.2. The histogram below shows the distribution of the number of hours per week students in a class spend on social media.



- How many students are represented in total?
 - What is the relative frequency of the 10–15 hours bin?
 - Is the distribution symmetric or skewed? Justify your answer in one sentence.
- 2.5.3. Describe the difference between a bar chart and a histogram. Give one example of data that would be displayed with a bar chart and one that would be displayed with a histogram.

2.6 Probability Distributions

A frequency table summarizes what was observed. A probability distribution goes one step further: it assigns a probability to each possible outcome, whether those probabilities come from a mathematical model or from observed data. This idea is central to probability and statistics because it describes not just what did happen, but the structure of what can happen.

Definition. *Probability distribution.* A **probability distribution** is a complete assignment of probabilities to all possible outcomes of an observation. It satisfies two conditions.

- (i) Every probability is a number between 0 and 1.
- (ii) The probabilities of all outcomes sum to exactly 1.

When probabilities are estimated from observed counts, the result is an **empirical** probability distribution. When probabilities are derived from an idealized mathematical model, the result is a **theoretical** probability distribution. Both viewpoints are useful, and much of statistics involves comparing the two.

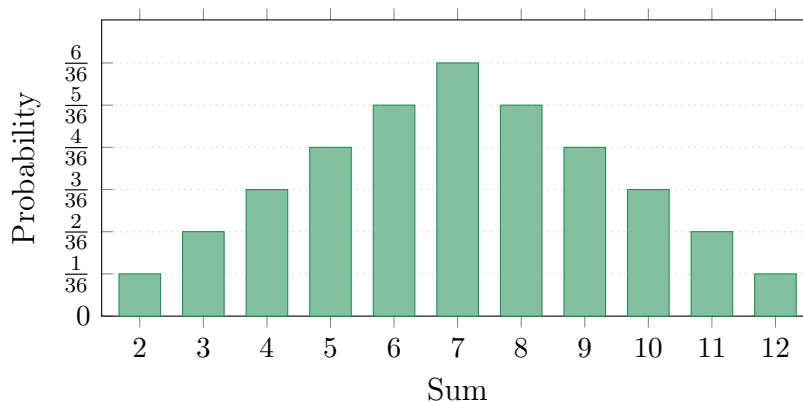
Example 2.6.1. (*Theoretical distribution*) A fair coin has two outcomes: heads (H) and tails (T). Each outcome has probability $\frac{1}{2}$.

Outcome	Probability
H	1/2
T	1/2
Total	1

This is a **uniform distribution**: every outcome has equal probability.

Example 2.6.2. (*Sum of two dice*) When two fair six-sided dice are rolled, the possible sums range from 2 to 12. The outcomes are not equally likely because there are more ways to produce a sum of 7 than a sum of 2.

Sum	Ways to achieve it	Probability
2	1	1/36
3	2	2/36
4	3	3/36
5	4	4/36
6	5	5/36
7	6	6/36
8	5	5/36
9	4	4/36
10	3	3/36
11	2	2/36
12	1	1/36
Total	36	1



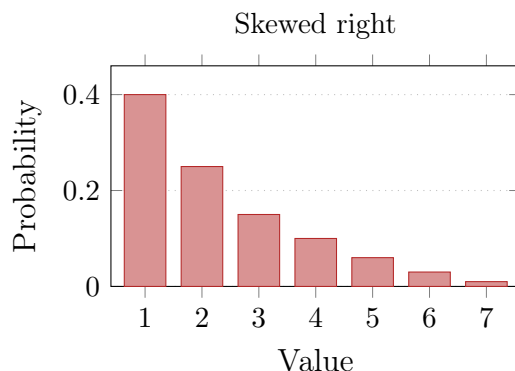
The distribution is symmetric about the sum of 7. The sum of 7 is the most probable, with probability $\frac{6}{36} = \frac{1}{6}$.

Shape of a Distribution

Three features describe the shape of a distribution.

1. **Center.** The value around which the distribution is concentrated. The most probable outcome is found at the tallest bar.

2. **Spread.** How widely the values are distributed. A distribution with bars spread across many values has greater variability than one concentrated in a small range.
3. **Shape.** A distribution is **symmetric** when the left and right sides are mirror images of each other. A distribution is **skewed right** when it has a longer tail on the right side. A distribution is **skewed left** when it has a longer tail on the left.



Income distributions and response times are typically skewed right: most values are modest, and a small number are very large.

Exercises 2.6

2.6.1. Determine whether each description corresponds to a valid probability distribution. If not, explain what is wrong.

- (a) Outcomes A, B, C with probabilities 0.3, 0.5, 0.2.
- (b) Outcomes A, B, C with probabilities 0.4, 0.4, 0.4.
- (c) Outcomes A, B with probabilities 0.7 and -0.3 .
- (d) Outcomes A, B, C, D with probabilities 0.25, 0.25, 0.25, 0.25.

2.6.2. A bag contains 3 red, 4 blue, and 5 green chips. One chip is drawn at random.

- (a) Construct the probability distribution.
- (b) Draw a bar chart of the distribution.
- (c) What is $P(\text{not blue})$?

2.6.3. The two-dice sum distribution from Example 2.6.2 is symmetric about 7.

- (a) Verify: what is $P(\text{sum} \leq 6)$?
- (b) What is $P(\text{sum} \geq 8)$?

- (c) Are the two probabilities equal? Explain why this is expected given the symmetry.

2.6.4. The following probability distribution describes the number of errors on a randomly selected document.

Errors	Probability
0	0.50
1	0.30
2	0.12
3	0.05
4	0.03

- (a) Verify that this is a valid distribution.
- (b) What is the probability of finding at least one error?
- (c) Sketch a histogram of this distribution. Is it symmetric or skewed? In which direction?

Chapter 2 Review

6. A quality control team inspects 50 products and records the number of defects per item: 0 defects (28 items), 1 defect (14 items), 2 defects (6 items), 3 defects (2 items).
- (a) Construct a frequency and relative frequency table.
- (b) Draw a histogram of the relative frequencies.
- (c) What is the probability that a randomly selected item has at least 1 defect?
- (d) Describe the shape of the distribution.
6. A survey of 60 commuters records travel time to work (in minutes): $[0, 15)$: 12 commuters, $[15, 30)$: 22, $[30, 45)$: 18, $[45, 60)$: 8.
- (a) Construct a relative frequency table.
- (b) Draw a histogram.
- (c) What is the probability that a randomly selected commuter travels between 15 and 45 minutes?

6. A spinner has four sections with probabilities $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{6}$, and p . Find p . Then find the probability of landing on one of the first two sections.
6. Explain in two complete sentences the difference between an empirical and a theoretical probability distribution. Give one example of each.
6. A class of 30 students is surveyed about how many hours of sleep they got the previous night. Results: 4 hours (2 students), 5 hours (4), 6 hours (8), 7 hours (10), 8 hours (5), 9 hours (1).
 - (a) Construct the relative frequency table.
 - (b) Draw a histogram.
 - (c) Describe the shape, center, and spread of the distribution.
 - (d) What is the probability that a randomly selected student slept fewer than 7 hours?
6. A school administrator claims the students in one advisory class are representative of the whole school. The class has 20 students, all seniors. The school has 400 students divided equally across four grade levels. Identify the population and sample, and explain in two sentences why this sample is biased.